HOTEL REVENUE PROTECTION

AI-Powered Booking Cancellation Predictor

Predict cancellation risk • Support smarter deposit decisions

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Dataset: Hotel Booking Demand | Model: Tuned Random Forest

The business problem

Why booking cancellation risk matters

THE CHALLENGE

Cancellations create uncertainty for hotel revenue and room planning.

A booking that looks secure today may not generate the expected stay revenue later.

THE OPPORTUNITY

Estimate risk early, then apply a proportionate deposit policy.

- Identify bookings with elevated cancellation probability
- Support consistent, data-informed deposit decisions
- Give staff a practical prediction at booking time

Project objective & workflow

From booking information to an actionable risk recommendation

01

Booking data

Guest, stay, channel and arrival details

→

02

ML pipeline

Preprocessing + trained classifier

→

03

Risk score

Cancellation probability

→

04

Policy output

No / Partial /Full Deposit

Target: is_canceled | Output: predicted probability + decision-support deposit tier

Dataset overview

Hotel Booking Demand — historical reservation records

119,390

Original records

Before cleaning

32

Original columns

Reservation and guest attributes

87,229

Final modeling rows

After cleaning and feature selection

23

Model features

Numeric, categorical and engineered

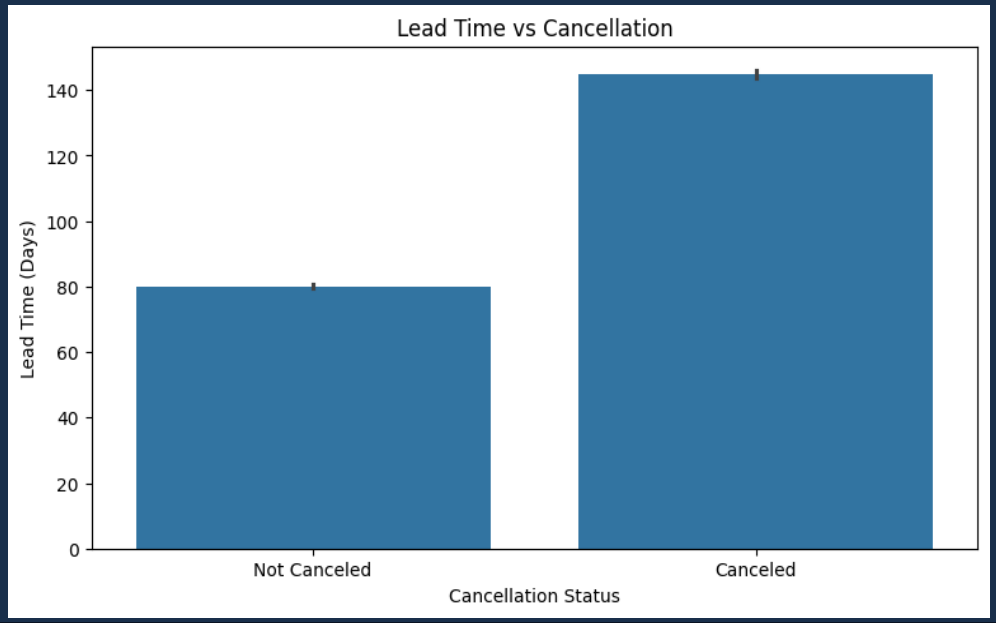
TARGET VARIABLE

is_canceled

0 = Not canceled

1 = Canceled

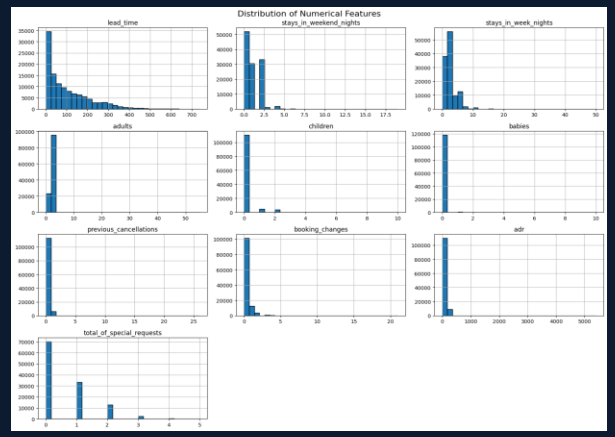
Source: Kaggle — Hotel Booking Demand (Jesse Mostipak).

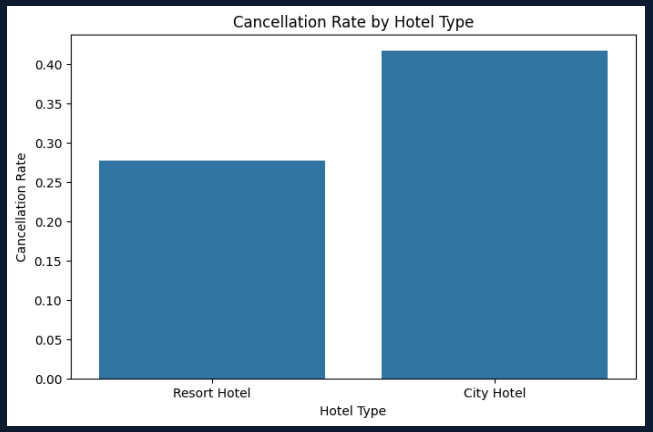


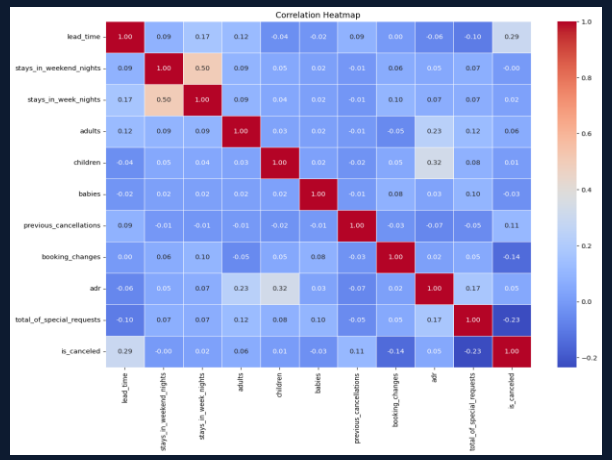
Exploratory Data Analysis (EDA)

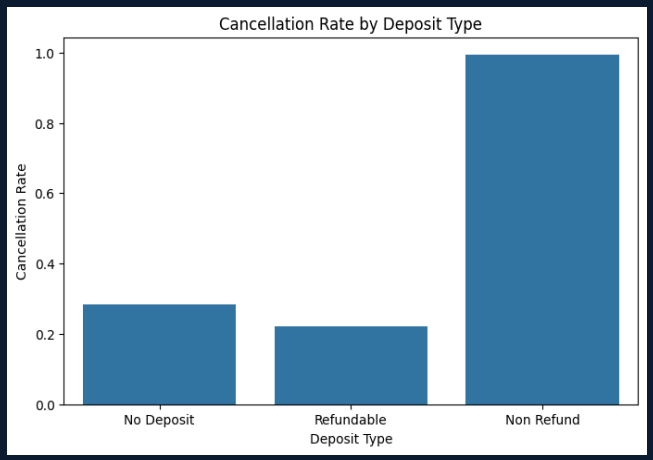
EDA insights

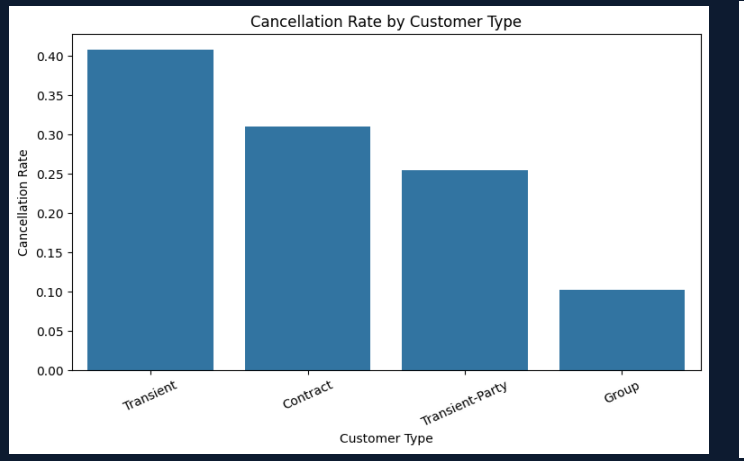
- Cancellation rates differ between City Hotel and Resort Hotel.
- Some market segments have higher cancellation rates.
- Longer lead times are associated with more cancellations.
- Cancellation rates vary across arrival months.
- Special requests or previous cancellations are associated with booking outcomes.

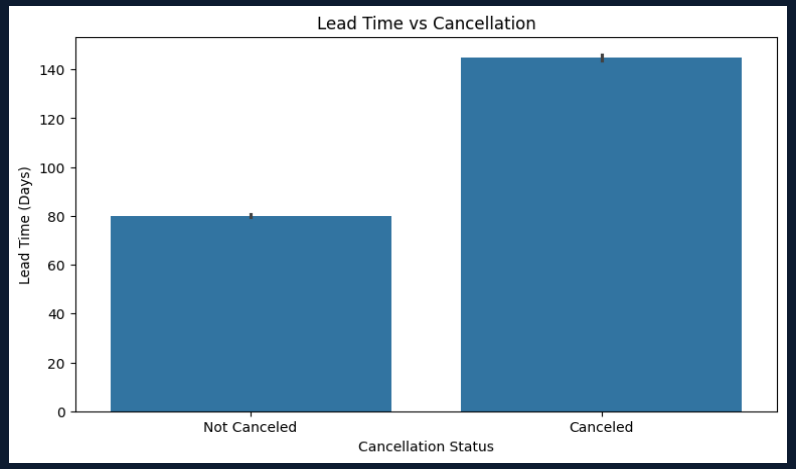


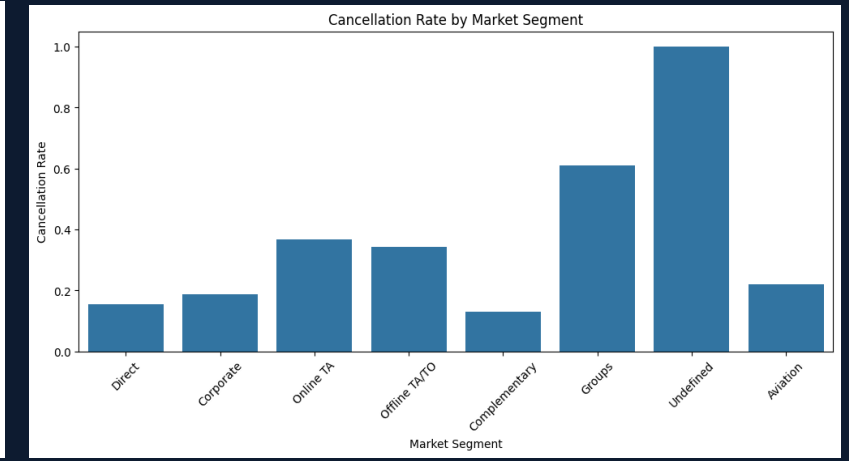












Data preparation & leakage control

Use only information available when a booking is made

CLEANING

- Removed 31,994 duplicate rows
- Removed 166 records with zero guests
- Removed one record with negative ADR
- Handled missing values inside the ML pipeline

AVOIDING DATA LEAKAGE

- Excluded reservation_status and reservation_status_date (post-outcome information)
- Excluded assigned_room_type, booking_changes and days_in_waiting_list (not known at initial booking)
- Excluded deposit_type to avoid feeding existing policy into the recommender
- Excluded highly incomplete/high-cardinality fields such as company and agent

Features & preprocessing pipeline

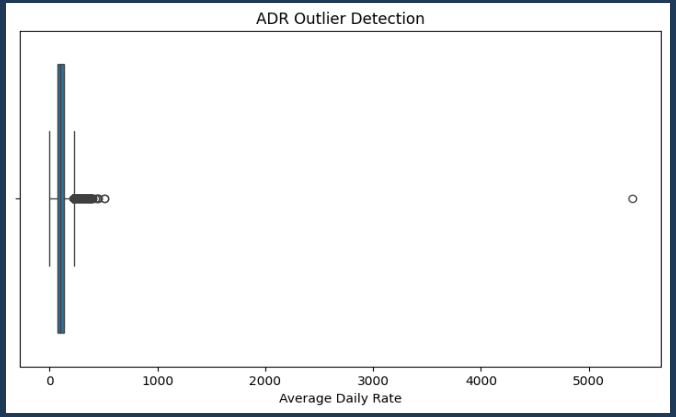
Preprocessing is fitted on training data through scikit-learn pipelines

FEATURE ENGINEERING

- total_nights = weekend + week nights
- total_guests = adults + children + babies
- arrival_day_of_week
- arrival_month_num

NUMERIC FEATURES

- Median imputation
- StandardScaler
- Examples: lead time, ADR, guests, nights, requests



CATEGORICAL FEATURES

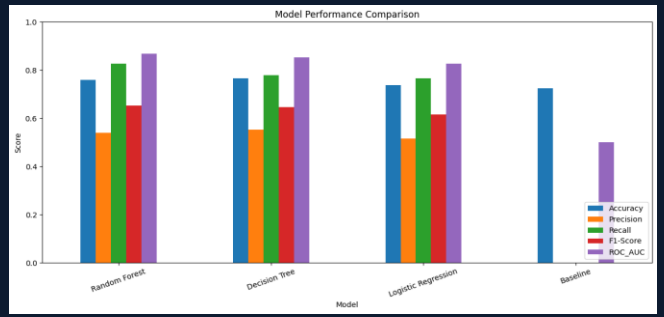
- Missing values → Unknown
- OneHotEncoder
- handle_unknown='ignore'
- Examples: hotel, meal, market segment, room type

Model development

Compare a baseline with three supervised classification algorithms

Dummy baseline	Reference performance	**0.500**	ROC-AUC
Logistic Regression	Linear classification	**0.827**	ROC-AUC
Decision Tree	Rule-based splits	**0.853**	ROC-AUC
Random Forest	Ensemble of decision trees	**0.869**	ROC-AUC

Validation: stratified 5-fold cross-validation on the training split.



Validation & hyperparameter tuning

Keep the test set untouched until final evaluation

80 / 20

Stratified train–test split

Random state = 42

5-fold

Cross-validation

Model comparison

10

Randomized search iterations

3-fold CV, ROC-AUC scoring

Best Random Forest configuration

150 trees • max_depth=None • max_features=0.5 • min_samples_split=5 • min_samples_leaf=5

Best cross-validation ROC-AUC: 0.8798

Final test-set performance

Tuned Random Forest evaluated on 17,446 unseen bookings

81.73%

Accuracy

64.30%

Precision — canceled

75.57%

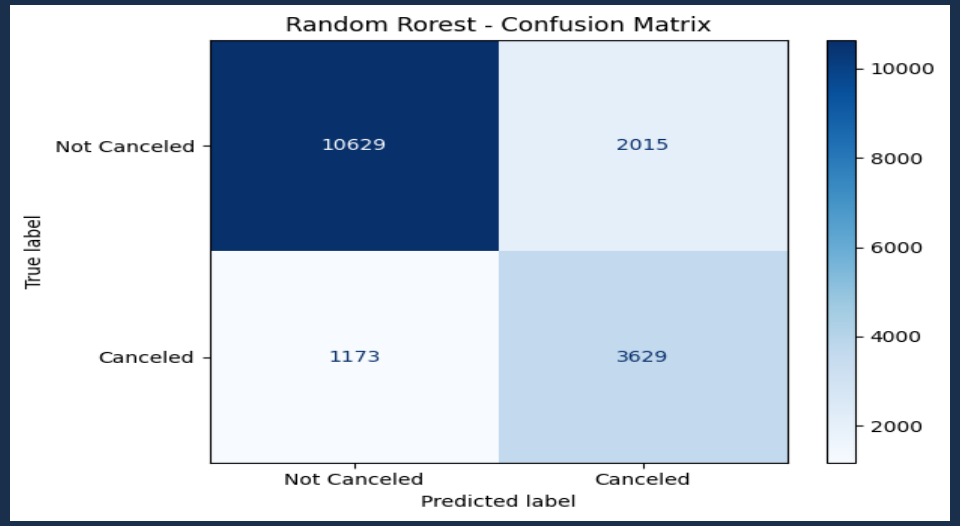
Recall — canceled

0.8897

ROC-AUC

Area under ROC curve

CONFUSION MATRIX



WHAT THIS MEANS

- Detected about 76% of actual cancellations
- About 64% of flagged bookings did cancel
- 1,173 cancellations missed; 2,015 non-cancellations flagged

From probability to deposit policy

Three risk bands translate model scores into recommendations

BELOW 30%

NO DEPOSIT

Lower predicted risk

30% TO <60%

PARTIAL DEPOSIT

Moderate predicted risk

60% OR ABOVE

FULL DEPOSIT

Higher predicted risk

Policy thresholds: 0.30 and 0.60 | Classification threshold: 0.50

Proposed decision-support bands; not proof that deposits prevent cancellations.

Deposit policy: observed test-set segments

Actual cancellation rates increased across predicted-risk groups

NO DEPOSIT

9,049

4.85% actual cancellations

8.72% avg predicted risk

PARTIAL DEPOSIT

3,871

31.23% actual cancellations

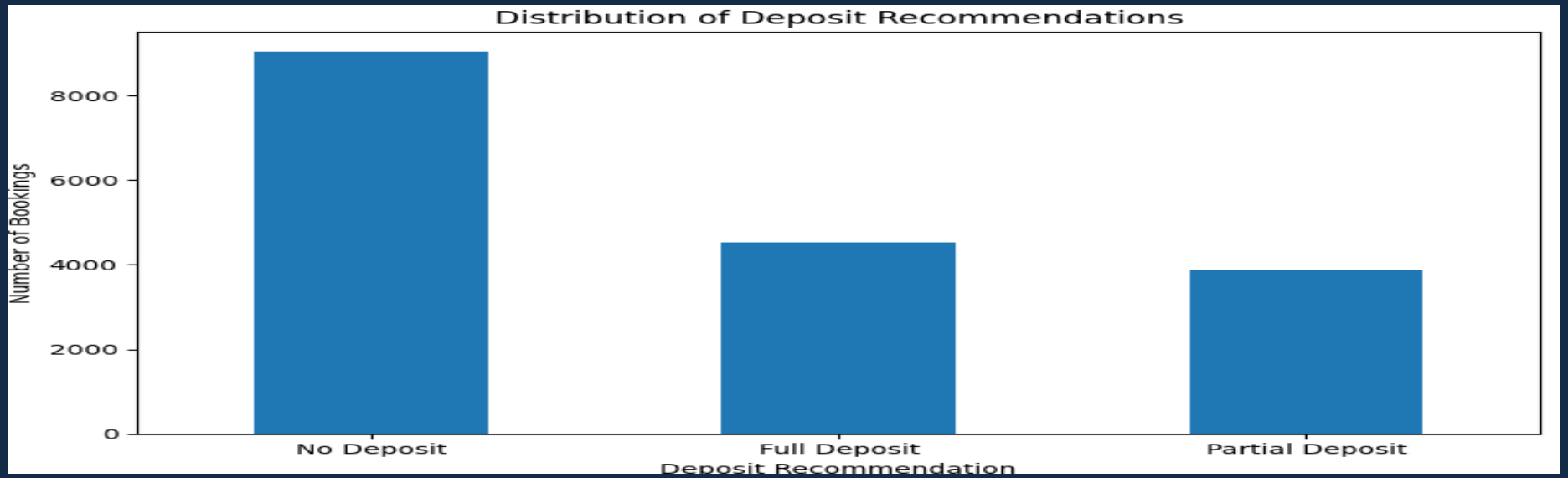
43.94% avg predicted risk

FULL DEPOSIT

4,526

69.69% actual cancellations

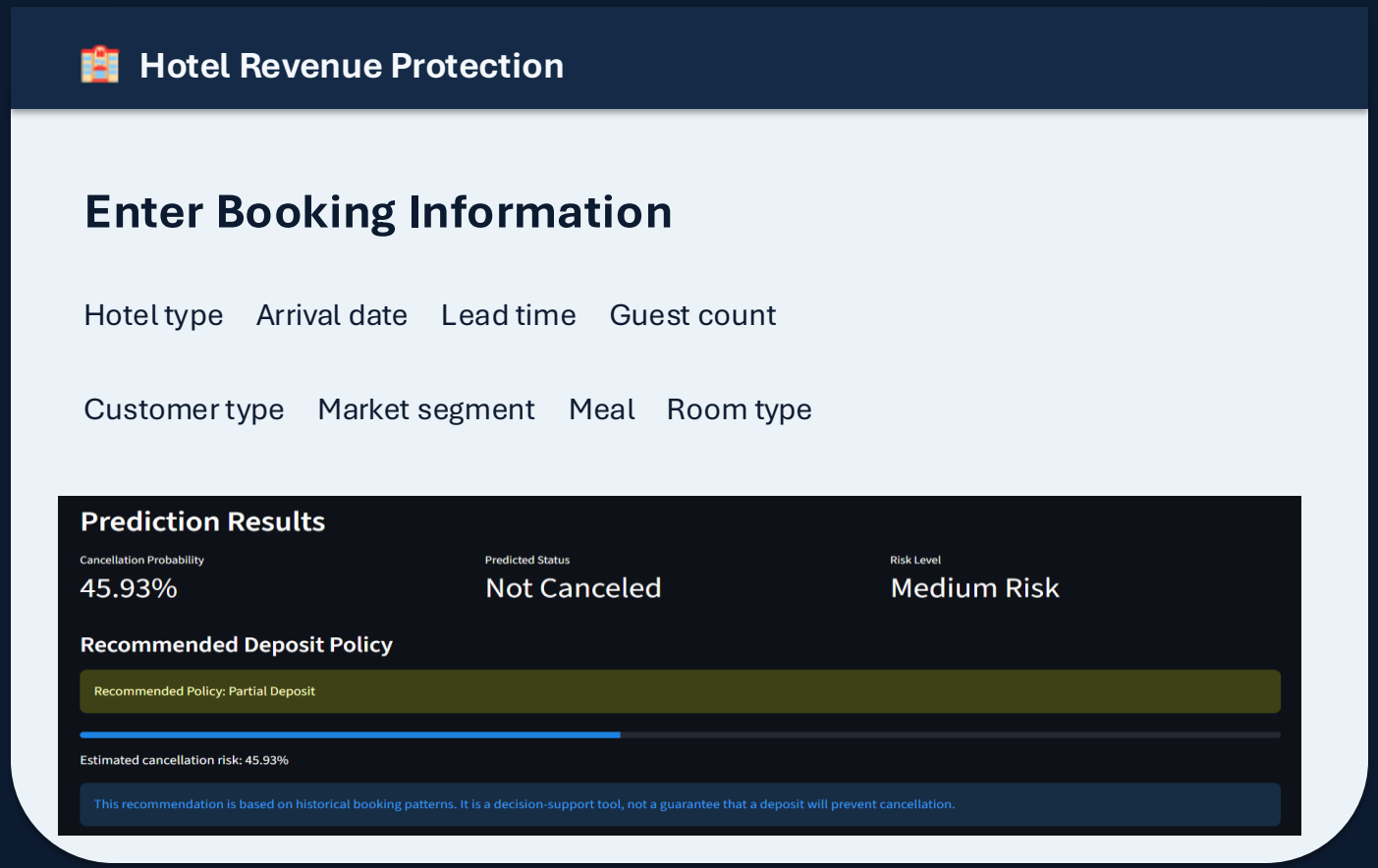
79.48% avg predicted risk



Segments show risk separation; they do not measure revenue saved or a causal deposit effect.

The deployed application

A user-facing interface for booking-level prediction



LIVE DEPLOYMENT

Streamlit Community Cloud

Public app URL

Hotel Revenue Protection · Streamlit

Limitations & future improvements

Responsible use and a roadmap beyond the prototype

LIMITATIONS

- Historical data may not reflect current travel patterns
- Predicted risk is not a guarantee of cancellation
- Dataset does not establish whether deposits cause fewer cancellations
- Performance may vary across hotels, markets and time periods

FUTURE WORK

- Monitor performance and retrain with recent booking data
- Calibrate probabilities and test policy thresholds with business costs
- Add revenue, occupancy and cancellation-cost analysis
- Evaluate segment-level performance and fairness
- Connect with hotel reservation systems

KEY TAKEAWAYS

A practical ML prototype for more informed booking decisions.

- A leakage-aware pipeline predicts booking cancellation risk.
- The tuned Random Forest achieved 0.8897 ROC-AUC on the held-out test set.
- Risk bands turn probability scores into No, Partial or Full Deposit recommendations.
- The Streamlit app makes the workflow accessible through a public web interface.

THANK YOU